One mark of an outstanding physicist is an ability to "smell" which experiments are to be trusted, and which not; which are to be worried about, and which ignored. As technology improves and the experiments are repeated over and over again, the truth ultimately becomes clear; but if one is trying to contribute to the progress of science, and if one wants to place one's own imprimatur on major discoveries, then one needs to divine early, not later, which experiments to trust.

Several outstanding physicists of the 1890s examined the Michelson—Morley experiment and concluded that the intimate details of the apparatus and the exquisite care with which it was executed made a strongly convincing case. This experiment "smells good," they decided; something might well be wrong with the foundations of Newtonian physics. By contrast, Heinrich Weber and most others were confident that, given time and further experimental effort, all would come out fine; Newtonian physics would triumph in the end, as it had so many times before. It would be inappropriate to even mention this experiment in one's university lectures; one should not mislead young minds.

The Irish physicist George F. Fitzgerald was the first to accept the Michelson—Morley experiment at face value and speculate about its implications. By comparing it with other experiments, he came to the radical conclusion that the fault lies in physicists' understanding of the concept of "length," and correspondingly there might be something wrong with Newton's concept of absolute space. In a short 1889 article in the American journal *Science*, he wrote in part:

> I have read with much interest Messrs. Michelson and Morley's wonderfully delicate experiment. . . . Their result seems opposed to other experiments. . . . I would suggest that almost the only hypothesis that can reconcile this opposition is that the length of material bodies changes, according as they are moving through the aether [through absolute space] or across it, by an amount depending on the square of the ratio of their velocities to that of light.

A tiny (five parts in a billion) contraction of length along the direction of the Earth's motion could, indeed, account for the null result of the Michelson–Morley experiment. But this required a repudiation of physicists' understanding of the behavior of matter: No known force could make moving objects contract along their direction of motion, not even by so minute an amount. If physicists understood correctly the nature of space and the nature of the molecular forces inside solid bodies, then uniformly